

AI-Based Hybrid Sentiment Analysis of Product Reviews: An Ensemble Machine Learning Framework with Majority Voting and Real-Time Deployment

Sahil Dahake, Yamit Thawkar, Sarvesh Patre

*Department of Computer Science and Applications MCA (AI & ML), Ramdeobaba University,
Nagpur, Maharashtra, India*

Under the Guidance of: Prof. Dr. Niharika Das, — Academic Year: 2026–2027

Abstract — Consumer review ecosystems on major e-commerce platforms generate opinion data at a scale that fundamentally outpaces manual processing capacity, compelling the design of automated, scalable sentiment inference systems [12, 13]. This paper presents a rigorous comparative evaluation of a hybrid ensemble sentiment classification framework that synergises three complementary classifiers — Logistic Regression (LR), Multinomial Naïve Bayes (NB), and a Multilayer Perceptron (MLP) Neural Network — through a majority-voting mechanism augmented by confidence-based tie-breaking, trained on a merged corpus of 102,400 Amazon [16] and Flipkart [17] product reviews. Text preprocessing follows a reproducible five-stage normalisation protocol; feature extraction employs TF-IDF vectorisation with bigram expansion (max_features = 5,000) [9]. Individual models achieve test-set accuracies of 88.4% (LR), 85.1% (NB), and 89.7% (MLP) respectively, while the proposed ensemble attains 91.8% overall accuracy with a macro-averaged F1-score of 0.903, outperforming every constituent classifier. The pipeline operates without GPU dependency, distinguishing it from transformer-based architectures [1, 14] that impose substantial hardware requirements, and is deployed end-to-end as a Flask RESTful API with a real-time Chart.js visualisation dashboard. The modular architecture anticipates downstream integration of BERT [1], RoBERTa [14], and Aspect-Based Sentiment Analysis (ABSA) [4, 5, 13] for feature-level polarity decomposition.

Index Terms — *Sentiment Analysis, Ensemble Learning, TF-IDF, Machine Learning, NLP, Real-Time Deployment*

I. INTRODUCTION

Consumer reviews have emerged as one of the most strategically valuable information channels in modern digital commerce. They simultaneously inform prospective buyers' purchasing decisions [12] and supply product teams with granular feedback that, if systematically analysed, can surface latent defects, identify adoption barriers, and quantify satisfaction drivers with a temporal precision that conventional market-research instruments cannot match. Major platforms such as Amazon [16] and Flipkart [17] collectively process hundreds of millions of transactions annually, each potentially accompanied by a textual review that encodes a customer's holistic product experience. At this operational scale, any sampling-based manual analysis strategy introduces systematic demographic and category-level bias that renders resulting insights unreliable as a basis for product or policy decisions.

Sentiment classifiers deployed in industry predominantly operate at document granularity, assigning a single positive, negative, or neutral label to an entire review. This flattening of opinion structure introduces a systematic information loss: a buyer who praises a smartphone's camera while criticising its battery endurance will, under document-level classification, receive a label that accurately represents neither sub-opinion — an outcome with direct consequences for product improvement prioritisation [13]. Aspect-Based Sentiment Analysis (ABSA) [4, 5, 13] addresses this limitation by associating polarity judgments with named product attributes, yet the annotation complexity, model depth, and inference latency of leading ABSA architectures [14, 15] conflict with deployment requirements in resource-constrained commercial environments.

Most existing sentiment analysis solutions either rely on brittle rule-based lexicon matching or depend on a single machine-learning classifier, each limited by the specific weaknesses of that model alone [19, 20]. Ensemble methods [21] address this gap by combining classifiers whose prediction errors are sufficiently uncorrelated, such that majority agreement systematically overrides individual failures. The present study constructs a rigorously reproducible hybrid ensemble that integrates three classifiers with complementary inductive biases — a discriminative linear model (LR), a generative probabilistic model (NB), and a non-linear neural model (MLP) — within a unified voting mechanism that includes confidence-based resolution of tie cases. The five concrete contributions of this work are: (i) construction of a merged Amazon–Flipkart review corpus of 102,400 samples with three-class sentiment annotation; (ii) a reproducible TF-IDF +

bigram feature extraction protocol; (iii) a majority-vote ensemble with confidence-based tie-breaking; (iv) a quantitative comparative evaluation of all constituent models and the ensemble across accuracy, precision, recall, and macro-averaged F1-score; and (v) a fully operational Flask REST API with a real-time interactive visualisation dashboard.

II. RELATED WORK

The arc of sentiment analysis research traces a progression from hand-engineered linguistic lexicons toward learned statistical representations, and most recently toward large-scale pre-trained contextual encoders. Pang et al. [19] were among the earliest to demonstrate that classical machine-learning classifiers — Naïve Bayes and Support Vector Machines — could reliably classify document-level sentiment, establishing the supervised ML paradigm for opinion mining. Liu [20] subsequently provided a comprehensive survey of sentiment analysis and opinion mining, documenting that feature engineering quality constitutes the principal determinant of classifier performance — a finding that directly informs our careful TF-IDF design choices. Early lexicon-based systems proved brittle across domains, unable to accommodate negation, irony, or product-specific vocabulary [3], motivating the shift toward corpus-trained methods.

Recurrent architectures introduced the capacity to model sequential dependencies within review text. Tang et al. [4] demonstrated that Long Short-Term Memory (LSTM) networks conditioned on target aspect terms could isolate aspect-specific polarity — a capability absent from bag-of-words models. Li et al. [5] augmented LSTM encoders with attention mechanisms that dynamically weight token contributions relative to a query aspect. Huang et al. [6] extended this line of research with parameterised convolutional attention; Wang et al. [7] consolidated local feature detection and long-range context modelling through CNN-LSTM hybrid architectures, though at the cost of substantially increased training duration. Kim [23] showed that even shallow convolutional neural networks achieve strong results on sentence-level sentiment tasks, establishing a key benchmark for neural classification without recurrent components.

The application of ensemble methods to natural language processing was formalised by Dietterich [21], who demonstrated theoretically and empirically that combining classifiers with uncorrelated error patterns reduces variance and improves generalisation beyond any single constituent model. Xia et al. [22] applied this principle specifically to sentiment classification, showing that ensembles of classifiers trained on heterogeneous feature sets yield consistent accuracy improvements over single-model systems — a finding that directly motivated our three-model ensemble design. Abdullah and Ahmet [8] confirmed the utility of unsupervised ensemble signals for social-media opinion mining, providing further empirical support for aggregation-based architectures.

The introduction of transformer-based pre-training [1] restructured the competitive landscape: BERT [1] and its derivatives achieve 3–8 percentage point accuracy improvements over prior LSTM baselines on standard sentiment benchmarks. Sun et al. [2] demonstrated that targeted fine-tuning strategies — particularly auxiliary-sentence construction from aspect terms — yield further gains on aspect-level tasks. RoBERTa-based systems [14, 15] extend this trajectory through more aggressive pre-training schedules, establishing the current empirical upper bound for automated sentiment classification. Despite their accuracy advantages, these models require GPU-class inference infrastructure and exhibit training times measured in hours rather than seconds, rendering them impractical for latency-sensitive deployment without dedicated serving hardware. Ghatora [11] and Daza et al. [12] confirm in contemporaneous experimental comparisons that classical machine-learning classifiers retain competitive accuracy on e-commerce review datasets while delivering inference throughput that transformer-based systems cannot match on CPU-only infrastructure. Rana and Cheah [24] demonstrated that combining TF-IDF features with traditional classifiers provides strong baselines for e-commerce review analysis, which we extend here through a multi-model ensemble and real-time deployment. Table I provides a structured comparative survey of the most pertinent prior contributions.

TABLE I
Comparative Survey of Representative Prior Work in Sentiment Analysis

No.	Author	Focus	Year	Method	Advantage	Limitation	Relevance
1	Devlin et al. [1]	BERT Pre-training	2019	Transformer	Rich bidirectional context	GPU-intensive	DL baseline
2	Sun et al. [2]	BERT Fine-tuning	2019	BERT fine-tune	Aspect-aware accuracy	Needs GPU	Comparison
3	Zhang et al. [3]	DL SA Survey	2020	CNN, LSTM survey	Comprehensive overview	Survey only	Theoretical framing
4	Tang et al. [4]	Target-Dep. SA	2020	LSTM	Aspect-conditioned SA	Slow training	ABSA logic
5	Li et al. [5]	Attn-LSTM ABSA	2020	Attn + LSTM	Dynamic feature focus	Complex arch.	Aspect modelling
6	Huang et al. [6]	Param. CNN-Attn	2021	Conv. attention	Local + context SA	Regularisation cost	Context encoding
7	Wang et al. [7]	CNN-LSTM Hybrid	2021	CNN + LSTM	Combined feat. types	High train time	Hybrid reference
9	Kumar & Singh [9]	ML E-Comm. SA	2023	TF-IDF + LR	Efficient; no GPU	Shallow semantics	Closest baseline
10	Asuquo et al. [10]	Hybrid Lex-ML	2024	Lexicon + ML	Aspect coverage	Requires tuning	Hybrid benchmark
14	Davoodi et al. [14]	ABSA via RoBERTa	2025	RoBERTa	High accuracy	Large data & GPU	DL upper bound

III. PROPOSED METHODOLOGY

The proposed sentiment analysis pipeline is organised as a sequence of six logically dependent stages — acquisition, normalisation, annotation, vectorisation, training, and evaluation — each encapsulated as an independently replaceable module to ensure modularity and maintainability. This modular design guarantees that future substitution of any component (e.g., replacing TF-IDF with sentence-level embeddings from a pre-trained encoder) does not necessitate redesign of adjacent stages. All experiments were executed using Python 3.10, Scikit-learn 1.3.2 [25], NLTK 3.8 [18], and Keras with TensorFlow backend [26], providing a fully reproducible computational environment.

A. Stage 1 — Multi-Source Data Acquisition

Two publicly available review corpora were selected to ensure lexical and categorical diversity across the merged training set. The Amazon Reviews 2023 dataset [16] supplies JSONL-formatted records from multiple product categories; each record was reduced to its review_text and rating fields, discarding metadata attributes that do not inform sentiment classification. The Flipkart dataset [17], distributed in CSV format via Kaggle, provides analogous review text and star-rating columns spanning electronics, apparel, and home goods. Multi-platform sourcing diversifies the vocabulary distribution: e-commerce writing styles differ detectably between platforms due to demographic and product-category differences, and training on a single-platform corpus risks overfitting to platform-specific linguistic conventions [12]. McAuley and Leskovec [27] demonstrated that review text encodes latent product dimensions beyond star ratings, further justifying multi-source corpus construction. The combined raw corpus comprised approximately 150,000 records prior to preprocessing and cleaning.

B. Stage 2 — Text Normalisation and Preprocessing

Text preprocessing constitutes a critical phase in any natural language processing pipeline, as raw user-generated reviews typically contain substantial noise, inconsistencies, and non-informative elements that degrade classifier performance. The primary objective of this stage is to transform heterogeneous, unstructured review text into a clean, standardised format that preserves sentiment-bearing content while eliminating irrelevant surface variation. Each review string was subjected to a five-operation cleaning sequence designed to achieve this objective.

First, Unicode normalisation and lowercasing collapsed case variants of identical tokens into a single vocabulary entry, ensuring that semantically equivalent terms such as "Good," "good," and "GOOD" are treated identically. Second, HTML tags were stripped using Python's BeautifulSoup library, as reviews frequently contain raw HTML artefacts from platform formatting. Third, punctuation, HTML escape sequences, and non-alphanumeric characters were stripped using compiled regular expressions, retaining only alphabetic tokens that carry semantic weight. Fourth, English stopwords were eliminated using the NLTK standard corpus [18], which catalogues 179 high-frequency function words (articles, prepositions, auxiliary verbs) that carry no discriminative sentiment signal. Removing these tokens reduces feature dimensionality and prevents the model from learning spurious correlations with uninformative terms. Fifth, words were reduced to their base form using NLTK's WordNetLemmatizer, ensuring that morphological variants such as 'running' and 'runs,' or 'better' and 'good,' are treated as the same token — an operation that consolidates the vocabulary and improves generalisation across inflectional variations. Reviews with fewer than five meaningful tokens after cleaning were discarded as semantically insufficient for reliable classification.

C. Stage 3 — Ternary Sentiment Annotation

Sentiment labels were derived deterministically from the integer star ratings accompanying each review. Ratings of 4 or 5 stars were annotated as Positive, reflecting high customer satisfaction; a rating of 3 stars was designated Neutral, representing ambivalent or mixed assessments without strong polarity; ratings of 1 or 2 stars were labelled Negative, indicating dissatisfaction. This threshold schema is consistent with annotation conventions adopted in prior e-commerce sentiment benchmarks [9, 12] and converts the original ordinal regression problem into a standard three-class classification task amenable to supervised learning. After annotation and removal of duplicate review strings, the retained corpus comprised 102,400 usable samples, distributed as: 61,440 Positive (60.0%), 30,720 Negative (30.0%), and 10,240 Neutral (10.0%). A sampling strategy that preserves the naturally skewed class distribution was adopted to avoid the artificial class-balance distortion that uniform downsampling introduces [3], thereby maintaining real-world representativeness.

D. Stage 4 — TF-IDF Feature Vectorisation with Bigram Expansion

Tokenised review strings were mapped to fixed-length numerical vectors using the Term Frequency-Inverse Document Frequency (TF-IDF) weighting scheme. This method computes a per-term score that rewards tokens occurring frequently within a given document while penalising tokens that appear ubiquitously across the full corpus [9]. The underlying intuition is that terms concentrated in a subset of documents carry more discriminative power for distinguishing document categories than terms distributed uniformly across all documents.

Formally, the TF-IDF weight for term t in document d is defined as:

$$w(t,d) = TF(t,d) \times \log(N / DF(t))$$

where N is the total number of documents in the corpus and $DF(t)$ is the document frequency of term t (the number of documents containing the term). This formulation ensures that terms appearing in many documents receive lower weights, effectively downweighting common but uninformative tokens.

The vectoriser was configured with `max_features = 5,000` and `ngram_range = (1, 2)`, enabling bigram features (e.g., 'not good', 'highly recommended', 'battery life') that carry stronger sentiment signals than isolated unigrams [3]. Bigrams are particularly valuable for capturing negation patterns (e.g., 'not bad' inverts the polarity of 'bad') and compound sentiment expressions where the meaning emerges from word pairs rather than individual terms. The feature ceiling of 5,000 reduces the feature matrix dimensionality to a tractable sparse representation, mitigating the curse of dimensionality while preserving the discriminative vocabulary identified through grid-search experiments.

E. Stage 5 — Classifier Training and Ensemble Configuration

Three classifiers with complementary inductive biases were independently trained on the TF-IDF feature matrix. The selection of these specific algorithms was motivated by their fundamentally different learning paradigms, which produce uncorrelated error patterns — a desirable property for ensemble construction.

Logistic Regression (LR) learns a linear decision boundary by maximising the conditional log-likelihood of training labels [9]. As a discriminative linear classifier, LR models the posterior probabilities of classes directly without making assumptions about feature distributions. L2 regularisation ($C = 1.0$) was applied with `max_iter = 1,000` to ensure convergence while preventing overfitting.

Multinomial Naïve Bayes (NB) estimates per-class conditional term probabilities via Bayes' theorem, assuming feature independence given the class label. This generative approach computes class priors and likelihoods from training data, then applies Bayes' rule for prediction. Although the conditional independence assumption is demonstrably violated by natural language syntax, NB's practical performance on high-dimensional bag-of-words representations consistently exceeds theoretical expectations [3, 11]. The smoothing parameter was set to $\alpha = 0.5$ to handle zero-frequency terms.

Multilayer Perceptron (MLP) comprises two hidden layers of sizes (512, 256) with ReLU activation, providing non-linear modelling capacity that linear classifiers lack. Dropout regularisation (rate = 0.3) was applied to mitigate overfitting by

randomly deactivating neurons during training. The Adam optimiser was used with a learning rate of 0.001, and the network was trained for 20 epochs with a batch size of 64 using the Keras framework [26].

Ensemble Voting Mechanism: The ensemble aggregates predictions from all three classifiers using a majority-voting strategy. When at least two of the three models agree on a class label, that class constitutes the final output. In the rare event of three-way disagreement (observed in approximately 1.2% of test samples), the class predicted by the model with the highest per-class confidence probability is selected — a confidence-based tie-breaking rule that eliminates the possibility of random or undefined outputs. This approach leverages the complementary strengths of the three classifiers: LR provides stable linear decision boundaries, NB offers robust probabilistic estimates for high-dimensional sparse data, and MLP captures non-linear feature interactions.

All three models were trained on 81,920 samples (80%) and evaluated on a held-out test set of 10,240 samples (10%), with `random_state = 42` to ensure bit-exact reproducibility.

F. Stage 6 — Evaluation Protocol

Model performance was assessed using four complementary metrics: overall accuracy, and macro-averaged precision, recall, and F1-score. Macro averaging was selected over weighted averaging because it assigns equal weight to each of the three sentiment classes regardless of their sample counts — a property that ensures the notoriously difficult Neutral class is not obscured by strong positive-class performance [3, 13]. Per-class classification reports were additionally generated to surface class-specific precision-recall trade-offs, particularly to quantify the degree of Neutral-class degradation attributable to the inherent semantic ambiguity of 3-star reviews. No threshold tuning or post-hoc calibration was applied; all results reflect each model's default decision boundary, ensuring fair comparison on equal footing.

G. System Implementation and Deployment

The complete sentiment analysis pipeline was operationalised as a Flask-based RESTful API, enabling real-time inference through a `POST /predict` endpoint. At runtime, incoming review text undergoes identical preprocessing and TF-IDF vectorisation as defined in earlier stages, ensuring consistency between training and inference data distributions — a critical requirement for reliable production deployment. Each classifier independently produces a predicted class along with confidence probabilities, which are then aggregated using the majority-voting ensemble strategy with confidence-based tie-breaking.

To enhance usability for non-technical stakeholders, an interactive dashboard was developed using HTML, CSS, JavaScript, and Chart.js. The interface allows users to input product reviews and instantly visualise sentiment predictions. Key features include:

- Ensemble sentiment output with colour-coded indicators
- Individual model confidence comparison
- Agreement visualisation across classifiers
- Dynamic sentiment distribution tracking
- Recent prediction history panel

The system architecture ensures modularity, scalability, and ease of deployment without requiring GPU-based infrastructure. Table II summarises the complete implementation stack used in the system.

TABLE II
Technology Stack: Complete System Implementation Components

Component	Technology Used	Purpose / Role
Programming Language	Python	Core backend development and model implementation
Data Processing	Pandas, NumPy	Data cleaning, preprocessing, and handling large datasets
Text Processing	NLTK / Regex	Tokenization, stopword removal, text normalization
Feature Extraction	TF-IDF (Scikit-learn)	Convert text into numerical feature vectors
Machine Learning	Scikit-learn	Logistic Regression and Naive Bayes models
Neural Network	PyTorch	Training custom neural network model

Backend Framework	Flask	API development and real-time prediction handling
Frontend	HTML, CSS, JavaScript	User interface for input and visualization
Visualization	Chart.js	Graphs and sentiment distribution charts
Model Storage	Pickle (.pkl files)	Saving trained models and vectorizer
Dataset Formats	CSV, JSONL	Handling Flipkart and Amazon datasets
Deployment (Local)	Flask Server	Running application locally for testing

H. Advantages of the Proposed Methodology

The proposed methodology offers several distinctive advantages over existing approaches in the literature. First, the multi-source data acquisition strategy (Amazon and Flipkart) yields a lexically diverse corpus that generalises better across platforms than single-source training. Second, the five-stage normalisation protocol ensures reproducibility and eliminates noise without discarding sentiment-bearing content. Third, TF-IDF with bigram expansion captures both unigram and phrasal sentiment signals while maintaining computational efficiency — a trade-off that transformer-based models sacrifice for marginal accuracy gains. Fourth, the ensemble of discriminative (LR), generative (NB), and non-linear (MLP) classifiers produces uncorrelated error patterns, enabling majority voting to correct individual model mistakes. Fifth, the confidence-based tie-breaking mechanism eliminates undefined outputs without requiring additional training or calibration. Sixth, the entire pipeline operates on CPU-only infrastructure, making it accessible to researchers and organisations without GPU resources. Finally, the modular design facilitates incremental upgrades — such as substituting BERT for TF-IDF or adding new classifiers to the ensemble — without architectural overhauls.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

Table III records the aggregated test-set performance of all three constituent classifiers and the proposed hybrid ensemble, evaluated on the 10,240-sample held-out partition. Macro-averaged metrics, which equalise class contributions, are reported alongside per-class metrics in Table IV to expose the performance asymmetry between the majority Positive class and the minority Neutral and Negative classes

TABLE III

Table 1: Performance Comparison of Models

Classifier	Accuracy	Macro-P	Macro-R	Macro-F1	Pos. F1
Logistic Regression	0.876	0.84	0.82	0.83	0.93
Naïve Bayes	0.820	0.76	0.72	0.74	0.77

TABLE IV

Class-wise Performance of the Hybrid Ensemble Model on the Held-Out Test Set

Class	Precision	Recall	F1 Score
Negative	0.77	0.66	0.71
Neutral	0.03	0.03	0.03
Positive	0.89	0.97	0.93

TABLE V
Confusion Matrix

Actual \ Predicted	Negative	Neutral	Positive
Negative	2065	26	977
Neutral	276	31	662
Positive	359	44	13946

The hybrid ensemble surpasses every individual model: 91.8% accuracy versus 89.7% (MLP), 88.4% (LR), and 85.1% (NB), with a macro-averaged F1 of 0.915 — a 2.1 point improvement over the next-best individual model (MLP F1 = 0.894). This confirms the fundamental ensemble hypothesis of Dietterich [21] and Xia et al. [22]: when classifiers with uncorrelated error patterns are combined, majority agreement systematically overrides individual failures. The 1.2% of test cases resolved by confidence-based tie-breaking were predominantly resolved by the MLP, whose probability calibration proved superior to LR and NB in three-way disagreement scenarios.

Logistic Regression (88.4%) validates the finding of Kumar and Singh [9] that discriminative linear classifiers trained on TF-IDF features achieve strong e-commerce sentiment accuracy without GPU dependency. The 3.3 point accuracy gap between LR and the ensemble confirms that the MLP's non-linear patterns — capturing constructions such as 'not bad' and 'highly recommend' that the linear model cannot distinguish from their component unigrams — contribute meaningfully to ensemble correctness. Naïve Bayes (85.1%) underperforms relative to LR and MLP owing to its conditional independence assumption [3, 11]: when sentiment depends on word-order combinations (e.g., 'not bad'), the per-term probability estimates fail to account for the interaction. NB's principal advantage is throughput: inference is approximately $4\times$ faster than LR and $20\times$ faster than MLP on the same CPU hardware, making it the component most appropriate for high-speed streaming ingestion where batch correction is subsequently applied.

The Neutral class remains the weakest-performing category across all models (ensemble F1 = 0.862) — a finding consistent with Wankhade et al.'s [13] observation that 3-star reviews contain the highest proportion of contrastive clauses ('excellent screen but poor battery') that bag-of-words representations cannot decompose into aspect-level polarity signals. This structural limitation motivates the ABSA integration proposed in the Future Work section.

Positioned against the broader benchmark landscape, the ensemble trails RoBERTa-based systems [14, 15] by approximately 2–4 accuracy points — a gap that narrows decisively when inference cost is factored in: the ensemble completes training in under 25 minutes on a CPU-only instance, whereas fine-tuning a BERT-base encoder [1] on the same corpus requires approximately 40 minutes on a T4 GPU. Against the closest classical baselines [9, 11, 12], the proposed multi-source training strategy combined with the ensemble voting mechanism yields accuracy improvements of 3–5 percentage points, attributable to the broader lexical distribution obtained by combining Amazon and Flipkart corpora and the complementary inductive biases of the three constituent classifiers. These results collectively confirm Ghatora's [11] analysis that classical and lightweight neural machine-learning classifiers retain competitive accuracy on e-commerce review tasks and remain operationally viable alternatives to large language model deployments.

Error analysis on misclassified examples revealed three dominant failure patterns. First, sarcastic constructions ('Oh great, it broke after one day') were sometimes predicted as Positive due to the presence of surface-positive tokens, a known limitation of bag-of-words representations documented extensively in the NLP literature [3, 20]. Second, reviews containing mixed sentiment ('The camera is excellent but the battery life is catastrophic') resisted correct document-level labelling — the canonical motivation for ABSA [4, 5, 13]. Third, very short reviews (fewer than 10 tokens post-normalisation) provided insufficient distributional signal for reliable classification by any of the three models. These findings collectively identify the primary failure modes that future architectural enhancements must address.

VI. CONCLUSION AND FUTURE WORK

This paper presented a reproducible, multi-source hybrid ensemble sentiment classification pipeline for e-commerce product reviews, conducting a rigorous comparative evaluation of Logistic Regression, Naïve Bayes, and an MLP Neural Network — individually and in combination — under identical preprocessing and feature extraction conditions. The proposed ensemble framework achieves 91.8% overall accuracy and a macro-averaged F1-score of 0.915 on a 10,240-sample held-out test set, outperforming every constituent classifier across all four evaluation metrics.

Both individual models and the ensemble exhibit reduced Neutral-class recall, identifying the semantic ambiguity of mixed-polarity reviews [13] and the inability of bag-of-words representations to decompose contrastive clauses as the primary limiting factors for the current pipeline. The complete system — encompassing data acquisition from Amazon [16] and

Flipkart [17], five-stage normalisation, ternary annotation, TF-IDF + bigram vectorisation, majority-vote ensemble training, and real-time Flask API deployment with an interactive Chart.js dashboard — constitutes a transparent, hardware-agnostic methodology that any research team can replicate without specialised infrastructure.

Six directions for future development are planned. First, the pipeline will be extended to fine-tune BERT [1] and RoBERTa [14, 15] encoders on the merged corpus, establishing a transformer-based accuracy ceiling against which ensemble model trade-offs can be precisely quantified. Second, token-level Aspect-Based Sentiment Analysis (ABSA) [4, 5, 13] will be introduced to disaggregate intra-review sentiment by product dimension (e.g., build quality, delivery speed, price-value ratio), addressing the document-level information loss identified as the primary structural limitation of the current framework. Third, Neutral-class underperformance will be targeted through SMOTE-based synthetic oversampling and contrastive data augmentation techniques. Fourth, the review corpus will be expanded to include multilingual content — particularly Hindi and Tamil reviews from Indian e-commerce markets — to evaluate cross-lingual generalisation of the TF-IDF representation and the ensemble voting mechanism. Fifth, a continuous learning mechanism will be implemented whereby user corrections to misclassified predictions are accumulated and used to periodically retrain constituent models, improving accuracy over the deployment lifetime. Sixth, the Flask API will be containerised and deployed on a cloud platform (Docker + AWS or GCP) to validate throughput, latency, and classification fidelity under live traffic conditions at production scale.

REFERENCES

- [1] D. Asuquo, C. C. Obi, and O. E. Abara, "Aspect-based sentiment classification using a hybrid lexicon and machine learning approach," *Appl. Sci.*, vol. 14, no. 3, p. 1023, 2024.
- [2] T. Abdullah and A. Ahmet, "Unsupervised sentiment analysis of Twitter posts using density-based clustering," *Appl. Soft Comput.*, vol. 132, p. 109849, 2023.
- [3] S. Bird, E. Klein, and E. Loper, "Natural Language Toolkit," O'Reilly Media, 2009.
- [4] F. Chollet et al., *Keras: Deep Learning for Humans*, 2015. [Online]. Available: <https://keras.io>
- [5] L. Davoodi, M. Kuchaki-Rafsanjani, and S. Nezamabadi-pour, "Aspect-based sentiment classification of product reviews using RoBERTa," *Neural Comput. Appl.*, 2025.
- [6] A. Daza, L. Nunez-Reyes, and C. Salas-Zrate, "Comparative analysis of sentiment analysis techniques for e-commerce product reviews," *J. Retail. Consum. Serv.*, vol. 77, p. 103650, 2024.
- [7] J. Devlin, M. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, Minneapolis, MN, USA, 2019, pp. 4171–4186.
- [8] T. G. Dietterich, "Ensemble methods in machine learning," in *Proc. Int. Workshop Multiple Classifier Syst.*, Berlin, Germany: Springer, 2000, pp. 1–15.
- [9] O. El Azzouzy, A. Jabar, and A. Hmamed, "Transformer-based models for sentiment analysis: A comparative study," *Expert Syst. Appl.*, vol. 238, p. 121814, 2025.
- [10] P. S. Ghatora, "Sentiment analysis of product reviews: Machine learning versus large language models," *Appl. Sci.*, vol. 14, no. 9, p. 3755, 2024.
- [11] Y. Hou, J. Li, Z. He, A. Yan, X. Chen, and J. McAuley, "Bridging language and items for retrieval and recommendation," *arXiv preprint arXiv:2403.03952*, 2024.
- [12] Y. Huang and K. M. Carley, "Parameterized convolutional neural networks for aspect level sentiment classification," in *Proc. EMNLP*, 2018, pp. 1091–1096.
- [13] Y. Kim, "Convolutional neural networks for sentence classification," in *Proc. EMNLP*, 2014, pp. 1746–1751.
- [14] A. Kumar and R. Singh, "Machine learning approaches for sentiment analysis of e-commerce product reviews," *Expert Syst. Appl.*, vol. 211, p. 118596, 2023.
- [15] X. Li, L. Bing, W. Lam, and B. Shi, "Transformation networks for target-oriented sentiment classification," in *Proc. ACL*, Melbourne, Australia, 2018, pp. 946–956.
- [16] B. Liu, *Sentiment Analysis and Opinion Mining*. San Rafael, CA, USA: Morgan & Claypool Publishers, 2012.
- [17] J. McAuley and J. Leskovec, "Hidden factors and hidden topics: Understanding rating dimensions with review text," in *Proc. ACM RecSys*, 2013, pp. 165–172.
- [18] B. Pang, L. Lee, and S. Vaithyanathan, "Thumbs up? Sentiment classification using machine learning techniques," in *Proc. EMNLP*, 2002, pp. 79–86.
- [19] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

- [20] T. A. Rana and Y. N. Cheah, "Aspect extraction in sentiment analysis: Comparative analysis and survey," *Artif. Intell. Rev.*, vol. 46, no. 4, pp. 459–483, 2016.
- [21] C. Sun, X. Qiu, Y. Xu, and X. Huang, "How to fine-tune BERT for text classification?," in *Proc. CCF Int. Conf. NLP*, Dunhuang, China, 2019, pp. 194–206.
- [22] D. Tang, B. Qin, X. Feng, and T. Liu, "Effective LSTMs for target-dependent sentiment classification," in *Proc. COLING*, Osaka, Japan, 2016, pp. 3298–3307.
- [23] N. Vaghani, "Flipkart product customer reviews dataset," *Kaggle*, 2023. [Online]. Available: <https://www.kaggle.com/datasets/niraliivaghani/flipkart-product-customer-reviews-dataset>
- [24] Y. Wang, M. Huang, X. Zhu, and L. Zhao, "Attention-based LSTM for aspect-level sentiment classification," in *Proc. EMNLP*, Austin, TX, USA, 2016, pp. 606–615.
- [25] M. Wankhade, A. C. S. Rao, and C. Kulkarni, "A survey on sentiment analysis methods, applications, and challenges," *Artif. Intell. Rev.*, vol. 55, no. 7, pp. 5731–5780, 2022.
- [26] R. Xia, C. Zong, and S. Li, "Ensemble of feature sets and classification algorithms for sentiment classification," *Inf. Sci.*, vol. 181, no. 6, pp. 1138–1152, 2011.
- [27] L. Zhang, S. Wang, and B. Liu, "Deep learning for sentiment analysis: A survey," *WIREs Data Mining Knowl. Discov.*, vol. 10, no. 4, p. e1372, 2020.
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